QIMIN FENG

Mechanical Engineering Ph.D. Candidate | Robotics Systems · Embedded Control · Mechatronics

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OBJECTIVE

> Mechanical Engineering Ph.D. candidate specializing in underwater robotic systems, embedded real-time control, and mechatronic design. Delivered a free-swimming manta robot from CAD to validation, first-authored an IROS 2026 paper on soft-robotic propulsion. Seeking full-time roles in robotics systems engineering, embedded control, or mechatronics.

EDUCATION

Iowa State University | Ph.D. Candidate, Mechanical Engineering — Jan. 2023 - Expected Dec 2027

GPA: 3.97 / 4.00 | Research: Bio-inspired underwater robotics, experimental fluid mechanics, and flow-field sensing and reconstruction

Zhejiang University | B.Eng., Agricultural and Biological Systems Engineering — Sept. 2018 - June 2022

TECHNICAL SKILLS

Programming & Embedded: Python, PyTorch, MATLAB, Embedded C / C++, Arduino Framework, Raspberry Pi 5, ESP32-C6, Teensy, Linux / systemd, Git, FreeRTOS queues, and I2C peripheral drivers

Robotics & Controls: multirate real-time control, PID, state machines, trajectory generation, actuator coordination, safety interlocks, watchdogs, and fail-safe design

Communication & Hardware: CAN / SocketCAN / MIT Motor Protocol, ESP-NOW, PWM, serial, UDP / WebSocket; BLDC motors, servos, VCAs, IMUs, pressure / depth / force-torque / Hall sensors, and stereo cameras

Mechanical Design & Prototyping: SolidWorks, KiCad PCB design, soldering and circuit debugging, 3D printing, actuator / sensor packaging, and experimental fixtures

Perception: OpenCV stereo calibration, StereoSGBM, and multisensor synchronization and calibration

Experimental & Measurement: PIV, ATI Mini-40 force / torque sensing, NI DAQ, Simulink Real-Time, water-tunnel testing, and design of experiments

ROBOTICS SYSTEMS EXPERIENCE

Iowa State University · Zhong Lab | Graduate Research Assistant — Jan 2023 - Present

MARIS Underwater Robot | Systems Integration & Embedded Control — Mar. 2026 - Present

SolidWorks · Raspberry Pi 5 · Python · SocketCAN · OpenCV · PySide6 / QML

- Designed and integrated a free-swimming manta-inspired robot with two motors, two servos, an IMU, pressure sensing, and stereo cameras; advanced the system from mechanical design through prototype integration and 3 m pool validation.
- Implemented a 50 Hz supervisory controller and 200 Hz dual-motor position-command pipeline over CAN / MIT mode for two AK45-10 actuators, enabling synchronized flapping up to 6 Hz, differential steering, phase resynchronization, manual override, and smooth return-to-neutral.
- Developed a PySide6 / QML Steam Deck interface for gamepad command mapping, real-time state feedback, stereo video / depth display, E-Stop, and data logging / replay.
- Calibrated the stereo-camera system to <0.5 px reprojection error and implemented real-time StereoSGBM depth estimation and visualization.
- Engineered layered fail-safe behavior with link watchdogs, loss-of-command return-to-neutral, actuator enable gating, unified E-Stop / shutdown paths, and staged thermal protection; isolated faults across control, communication, and perception.

Constrained-Layer-Damping Soft Robotic Propulsor | Design & Hydrodynamic Validation — Jan. 2026 – Mar. 2026

IROS 2026 Accepted (First Author) · CLD · PIV · Experimental Design

- Designed and fabricated four coverage variants of a frequency-selective CLD soft propulsor using symmetric PLA / acrylic-foam / PET laminates.
- Planned and executed comparative testing across four configurations and St = 0.2–0.8, including 1–5 Hz structural-dynamics characterization, thrust measurement, phase-resolved PIV, and free-swimming validation.
- Increased thrust from 0.16 to 0.51 N (>200%) at St = 0.8, peak acceleration from 0.039 to 0.191 m/s² (~5×), and terminal speed from 0.105 to 0.291 m/s (~3×).

Modular Antagonistic VCA Joint | Embedded Control & Mechatronic Design — May. 2026 - Present

SolidWorks · ESP32-C6 · Embedded C / C++ · PID · ESP-NOW · KiCad

- Designed the antagonistic dual-VCA mechanism and sensor packaging in SolidWorks, and laid out the companion sensor PCB in KiCad; completed board assembly, soldering, and circuit debugging.
- Developed a Commander / Driver architecture across two ESP32-C6 modules with local sensing and actuation, acknowledged ESP-NOW commands, queued reception, and state feedback.
- Implemented and tuned 500 Hz PID angle control over ±45°; experimentally verified a linear relationship between PWM duty cycle and drive current for predictable open-loop actuation.
- Improved robustness with sensor filtering, dead-zone compensation, mechanical limits, encoder-fault latching, priority E-Stop, and automatic output shutdown on communication loss.

Flow-Sensing Platform | Pressure–PIV Sync & Data Acquisition — Apr. 2025 - Present

Teensy · TCA9548A · I2C · MS5803 · Python · PyTorch

- Built and debugged an MS5803-based five-channel pressure-acquisition system and used Teensy hardware triggering to synchronize 80 Hz pressure measurements with PIV on a shared time base.
- Collected 54 synchronized experiments spanning distinct flow and geometry conditions; developed Python pipelines for cleaning, per-channel zero-offset correction, synchronization checks, and pressure–PIV pairing.
- Designed a cross-attention neural decoder that reconstructs dense vorticity fields from five pressure inputs and achieved 92% flow-direction prediction accuracy; introduced learnable spatial tokens to overcome the sparse-sensor attention rank bottleneck, with physics-informed losses enforcing incompressibility.

SELECTED PUBLICATIONS

- **Feng, Q.**, Roberts, O. A., Zhong, Q. "Passive Phase-Oriented Impedance Shaping for Rapid Acceleration in Soft Robotic Swimmers." IROS 2026, accepted (first author); arXiv:2603.03537.
- **Feng, Q.**, Han, T., Zhong, Q. "Lift Reversal from Vortex-Surface Phase Coupling in a Heaving Foil near a Free Surface." Preprint available; submitted to Journal of Fluid Mechanics, 2025 (first author); arXiv:2512.12485.
- Jin, L., **Feng, Q.**, Gunnarson, P., Zhong, Q. "Write-Safe Flow Field Mapping under Ambiguous Onboard Sensing and Localization Drift." submitted to IEEE Robotics and Automation Letters, 2026 (second author).
- Zhang, Z., **Feng, Q.**, Zhu, Y., Zhong, Q. "Inertial Effects on the Mechanical Efficiency of a Semi-Passive Oscillating Hydrofoil Energy Harvester." Preprint available; submitted to Renewable Energy, 2026 (second author); arXiv:2606.11126.
- Zhu, Y., Liu, L., Han, T., **Feng, Q.**, et al. "Wavenumber Affects the Lift of Ray-Inspired Fins near a Substrate." J. R. Soc. Interface 22:20250276 (2025); DOI:10.1098/rsif.2025.0276.

HONORS & PRESENTATIONS

- Link Foundation Ocean Engineering & Instrumentation Fellowship | July 2024
- ISU CoMFRE Annual Meeting Presentation | Oct 2024
- APS Division of Fluid Dynamics Annual Meeting Presentation | Nov 2024